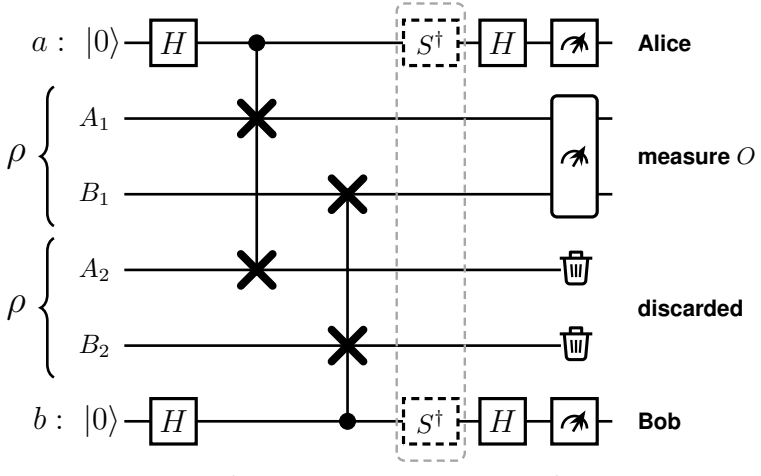


6Q LOCC — no $S^\dagger$: $C_{XX} = \langle X_a X_b \otimes O \rangle$; with $S^\dagger$: $C_{YY} = \langle Y_a Y_b \otimes O \rangle$
both runs: $\Delta(O) = C_{XX} - C_{YY} = \mathrm{Tr}(O\rho^2)$, $\langle O \rangle_{\mathrm{VD}} = \Delta(O)/\Delta(I)$